International Journal of Hospitality Management 132 (2026) 104356



Contents lists available at ScienceDirect

International Journal of Hospitality Management

journal homepage: www.elsevier.com/locate/ijhm



Full length article



Can nearby high-quality restaurants increase other tourism firms' performance? Evidence from hotels

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ARTICLE INFO

Keywords:
Restaurant cluster quality
Hotel performance
Agglomeration effects
Spillover effects
Inter-industry agglomerations

ABSTRACT

Co-locating phenomena have been extensively observed in restaurants and hotels, but prior studies have failed to reflect the reality and nature of agglomerations. From a tourism perspective, this study clarifies how various types of agglomerations can affect hotels and demonstrates that agglomeration externalities can occur between different industries. Using combined data from restaurants (customer reviews) and hotels (hotel performance), this study proves that the quality of co-located restaurants significantly influences hotel performance in terms of occupancy, room rate, and revenue. Highquality restaurants are an essential pull factor for tourists and cause positive agglomeration effects to neighboring hotels. Furthermore, this study confirms that the positive impact of the quality of a restaurant cluster becomes stronger in full-service hotels than in limited-service hotels, as full-service hotels are more relevant as coopetitors to restaurants.

1. Introduction

The excellent reputation of a restaurant can be an important pull factor for tourists (Daries et al., 2021; De Albuquerque Meneguel et al., 2019). In the recent digitalized era, this trend has become stronger since information on restaurants, including the menu, taste, pricing, service, and overall quality, is posted online and constantly evaluated via customers' photos and comments. For a few tourists, an exceptional restaurant experience may be a primary motivation (or pull) factor, but for many, it may be a secondary concern. In other words, although gourmet tourists choose destinations based mainly on the reputation of their target restaurants, most tourists consider other aspects of their destinations to be more important.

Despite different primary motivations, in both cases, gastronomy is an essential part of tourists' travel experiences (Sánchez-Cañizares and López-Guzmán, 2012) and significantly influences travel activities. In other words, even if food is not one's primary motivation for travel, memorable dining experiences will substantially increase the pleasure and value of traveling (Horng and Tsai, 2012; Kivela and Crotts, 2006; Sánchez-Cañizares and López-Guzmán, 2012). Therefore, the quality of nearby restaurants and alternative dining options should be an important consideration in terms of travel plans even if they are not gourmet tourists (Daries et al., 2021; De Albuquerque Meneguel et al., 2019).

Agglomeration theory focuses on a firm's unique co-location phenomenon as nearby competitors and/or similar businesses create benefits for each other (Mathias et al., 2021; McCann and Folta, 2008, 2009; McCann and Vroom, 2010; Wang et al., 2020). Much research has suggested that co-location benefits can be differentiated into externalities of production and demand (Marshall, 2009). Demand externalities are considered the most relevant to the hotel and restaurant industries since co-located hotels and restaurants benefit from increased demand (Canina et al., 2005; McCann and Folta, 2008). When restaurants are located together (e.g., food courts, Chinatown in New York City), consumers can reduce their time to decide on a restaurant (i.e., searching costs) by comparing their options, which increases the probability of a purchase (Kim et al., 2024). However, this traditional approach, which

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https://doi.org/10.1016/j.ijhm.2025.104356
Received 14 October 2024; Received in revised form 15 June 2025; Accepted 5 July 2025
Available online 15 July 2025
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Table 1
Summary of key studies on agglomeration: categories, benefits, findings, and theoretical relevance.

Categorization of Agglomeration	Types of Agglomeration Benefits		Key Findings / Argument	Example Studies	Gaps identified / Relevance to this study
Marshall's Agglomeration	Benefits from *specialization:* Localization Externalities	Supply-side benefits	• Firms benefit from sharing industry-specific inputs such as skilled labors, specialized suppliers, or unique resources (e.g., beach, cultural, coal, soil characteristics, etc.). • Externalities occur through knowledge, skill, or technology spillover (e.g., automotive, film, high-tech, hotel, steel, winery industries).	Kadiyali and Kosová (2013); Luo and Yang (2016); McCann and Folta (2008), (2009); Massimino et al. (2017); Rosenthal and Strange (2004); Woo and Mun (2020)	• Oversimplification of agglomeration. • Does not account for the complexity of real-world agglomerations where multiple forms (supply- and demand-side) coexist. • Hotel studies need to refocus from demand-side benefits to supply-side benefits (i.e., shared inputs, labor spillovers) • *This study combines it with Jacobs' externalities;* Restaurants and hotels rely on similar types of skilled labor and inputs. Co-location of good-quality restaurants and hotels enhances recruitment and performance through shared access to talent and supply chains.
		Demand-side benefits	• Firms benefit from access to heightened demand (i.e., co-locating shops allow customers to compare products more easily, saving search time, and increasing area demand) (e.g., food court at a local mall, restaurants along a street) • Frequently used by prior hotel agglomeration studies	Canina et al. (2005); Chung and Kalnins (2001); Kalnins and Chung (2004)	
Jacobs' Agglomeration	Benefits from *diversity:* Externalities	Urbanization	• Firms benefit from a variety of consumption and workforces, availability of infrastructure, innovation, and economic growth. • Externalities occur across industries (i.e., outside a firm's own industry) and increase with high industrial interaction and proximity	Duranton and Puga (2005); Glaeser et al. (1992); Henderson et al. (1995); Leonardi and Moretti (2023); McCann and Vroom (2010); Rosenthal and Strange (2003), (2004); Woo et al. (2019)	• Oversimplification of agglomeration. • Understudied in hospitality. • Justifies Marshall's demand-side agglomeration effects from other industries • *This study combines it with Marshall's externalities;* High-quality dining options improve destination attractiveness and creates consumption-driven externalities. Hotel guests benefit from nearby restaurants, especially where in-house options are limited.

typically identifies externalities in the same industry (i.e., direct competitors), has limitations in fully understanding the current and important phenomena in the tourism industry that show spillover effects from restaurants to hotels.

In the tourism context, we argue that good-quality restaurants have spillover effects or externalities on nearby hotels by attracting more tourists and locals. Furthermore, we suggest that the quality of co-located restaurants influences hotels in different ways (Leonardi and Moretti, 2023) due to different customer expectations and needs. In other words, spillovers from or externalities of nearby restaurants seem to exist and can also be explained using agglomeration economies. Therefore, we propose that externalities from different industries exist between nearby restaurants and hotels instead of focusing on one type of agglomeration effect (Barge-Gil et al., 2020; Kadiyali and Kosová, 2013). Furthermore, we suggest that these externalities originate primarily from restaurant clusters rather than hotel clusters. Unlike hotels (i.e., increase tourists through restaurants), restaurants may not need to be located near hotels to attract tourists as their major target customers are locals. In fact, many Michelin-starred restaurants are located in small towns and villages instead of big cities where hotels are highly located (Rickman, 2023; Sands, 2024). Instead, they would prefer locations where they can enjoy the advantages of quality local food sourcing with lower operating costs.

At a certain level of quality, restaurants act as a pull factor for enhancing tourist demands in all types of hotels, which positively impacts hotel performance. Although the factors of co-location are related to a firm's expectations of enjoying positive externalities, these benefits do not always occur in every case (Canina et al., 2005; Chung and Kalnins, 2001; Jung and Jang, 2019; Kalnins and Chung, 2004). Depending on the types of restaurants located nearby, hotels receive different benefits. If the quality of restaurants in the vicinity is outstanding, the effect on hotel performance differs between full- and limited-service hotels. This is because, unlike limited-service hotels, full-service hotels need to compete with neighboring restaurants in terms of food quality, services, and customer demands. If this competition improves the reputation and quality of hotel restaurants, intensified competition may be beneficial for full-service hotels. On the contrary, the high quality of neighboring restaurants could rather be detrimental to full-service hotels if customers could easily find better F&B offerings from nearby restaurants. However, in both cases, limited-service hotels do not need to compete with nearby restaurants in terms of the quality of F&B offerings. In this context, we posit that the impact of quality restaurants near a hotel differs substantially between full- and limited-service hotels, which haven't been covered by previous studies. Therefore, this study aims to identify how the quality of restaurants in the vicinity influences hotel performance and to compare the discrepancies between full- and limited-service hotels.

This study contributes to the agglomeration literature and tourism industry in three ways. A number of prior studies have investigated agglomeration effects using simple typologies (e.g., Marshall's demand-side agglomeration), but they fail to explain the reality and mechanisms of agglomeration (McCann and Folta, 2009). To fill these gaps, this study highlights that the tourism industry greatly influences the nature of diverse hospitality (i.e., restaurants and hotels) firms' agglomeration externalities, suggesting differences among types of agglomeration in the context of tourism. This study not only argues that hospitality firms cause complex and various agglomeration effects but also tests whether agglomeration effects can exist between different but associated firms.

Thus, this study first strengthens the agglomeration theory by providing evidence of the heterogeneity of agglomeration effects: agglomeration effects exist between different industries (i.e., inter-industry agglomeration effects). Second, based on the findings of previous agglomeration studies, this study focuses on the quality of nearby restaurants and tests how this affects the performance of full- and limited-service hotels differently. The findings are practically important for both the restaurant and hotel industries because both compete yet cooperate to achieve better business performance. Finally, this study uses longitudinal panel data and combines different data sources, customers' restaurant reviews, and hotel information to provide more robust implications for both agglomeration and tourism studies.

Existing agglomeration studies have relied heavily on firm-level information and tested agglomeration externalities in one type of industry due to the limited availability of data (Kim et al., 2024). Many tourism studies that have used customers' information have been limited to estimating customers' satisfaction or behavioral intentions (Bilgihan et al., 2018; Camilleri and Filieri, 2023), so it is difficult to gauge how much customers affect business performance. Methodologically, this study is robust in that the models trace the quality of all vicinity restaurants and examine their impacts on hotel performance over 11 years. In these models, the potential correlations or clusters among repeatedly observed firms over the years (restaurants are expected to be clustered in certain locations) are controlled by applying generalized estimating equations (Gardiner et al., 2009; Mun and Woo, 2021). In addition, this study properly addresses the potential endogeneity issues of restaurant clusters by using the length of review and one-year lagged length of review as instrumental variables. The test results of instrumental variables confirm the relevance of two-stage least squares (2SLS) panel regression models for this study (Wooldridge, 2016). Besides, customer big data allows this study to verify agglomeration effects between different industries, and firm-level data facilitates proving how customer satisfaction can significantly change firm performance.

2. Literature review

2.1. Types of agglomeration

Scholars have investigated agglomeration phenomena and found that firms develop clusters in different forms, and these clusters affect different externalities. While agglomeration effects have been researched from different perspectives (e.g., international business, strategic management, organizational ecology, and entrepreneurship), the two most identified types of agglomerations are Marshall's and Jacobs' agglomerations (McCann and Folta, 2009; Rosenthal and Strange, 2004).

Marshall's agglomerations focus on the spatial concentration of *related* firms in the same industry, such as automotive manufacturing in Detroit, the high-technology industry in Silicon Valley, the movie industry in Hollywood, and the clothing industry in shopping malls (Massimino et al., 2017; McCann and Folta, 2008). This agglomeration relates to *localization* or *specialization economics*: firms cluster together due to greater access to unique, rare, or specialized resources, such as input, labor, knowledge spillover (e.g., transport nodes, unique climates, cultural heritages, skilled workers, and star scientists), and demand (e.g., concentration of customers). Based on specialized resource type, Marshall's agglomeration distinguishes between supply-side and demand-side agglomerations. Supply-side agglomerations are "agglomerations of *producers of industrial goods*," and demand-side agglomerations are "agglomerations of *sellers of consumer goods and services*" (McCann and Folta, 2009, p. 364).

On the other hand, Jacobs' agglomeration is based on geographic concentrations of *unrelated* firms, and externalities are created from *urbanization economies*. For example, different types of firms aggregate in cities, and the diversity of products and services generates benefits across industries. Related studies on Jacobs' agglomeration focus on the infrastructure and consumption amenities of cities. These have shown consistent results, in that urbanized effects increase the birth and growth rates of different industries (Glaeser et al., 1992; Henderson et al., 1995) and encourage diverse employment (Henderson et al., 1995; Rosenthal and Strange, 2003). Unlike Marshall's externalities, which make assumptions regarding economic activities in one's own industry, Jacobs' externalities support the idea that economic activities occur outside of one's own industry due to easy accessibility and geographic closeness

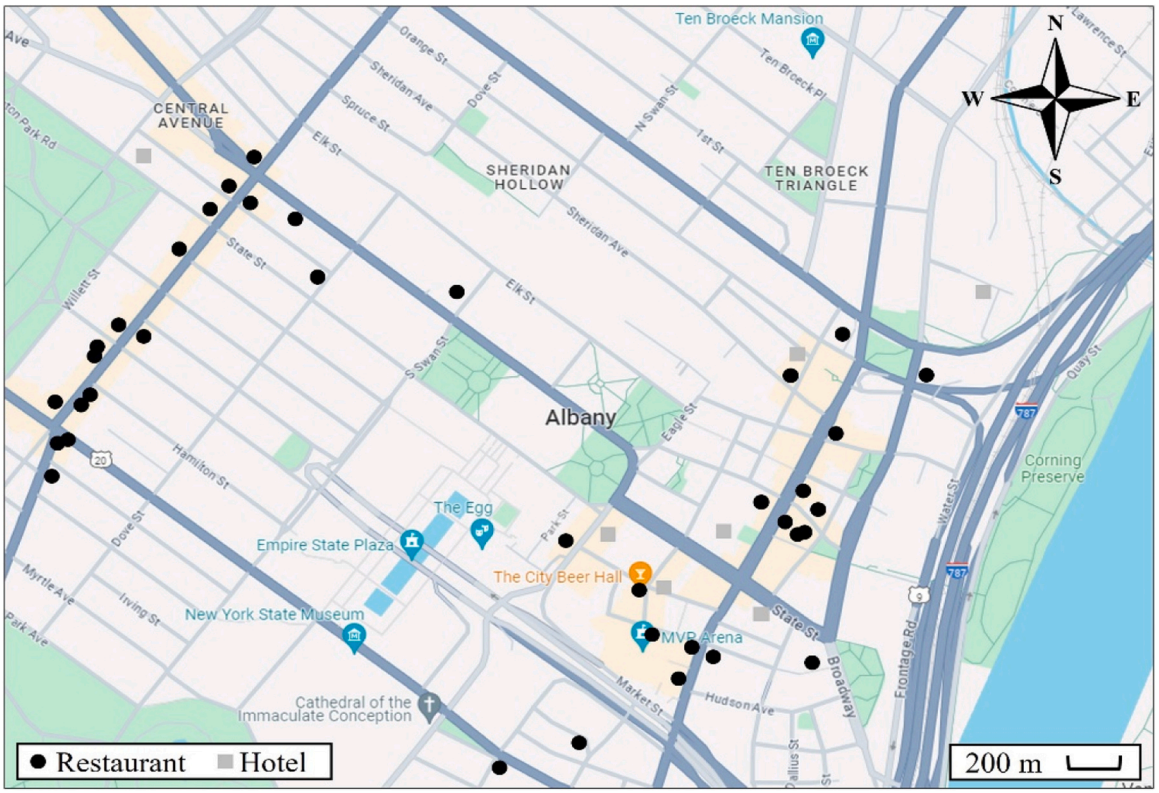


Fig. 1. Geographic distribution of restaurants and hotels in Albany (Source: Google Maps).

(Rosenthal and Strange, 2004).

Although Marshall's agglomeration is the most prevalent theoretical explanation for agglomeration studies, several researchers have pointed out that Marshall's theoretical assumptions are too simple to reflect the reality of clusters. Scholars have criticized that explaining only one type of agglomeration oversimplifies agglomeration phenomena (Eaton and Lipsey, 1979; Ellison and Glaeser, 1999), as different forms of agglomeration are interdependent and should be understood in a more complex manner (McCann and Folta, 2008, 2009). For example, employees of supply-side agglomerated firms (e.g., biotechnology firms) can generate demand for services or goods offered in demand-side agglomerations (e.g., auto dealers, restaurants, and cafes). In large cities or hub-and-spoke regions (i.e., urban economies), business service clusters, such as financial services and marketing and consulting firms, benefit not only from specialized input, labor, and knowledge spillover (i.e., supply-side benefits) but also from accessing market-leading customers (i.e., demand-side benefits) (Coe et al., 2020; Cook et al., 2004).

To fill these research gaps, this study focuses on hospitality firm agglomerations and explains, by applying both Marshall's and Jacobs' agglomeration effects, how clusters of unrelated or related firms (i.e., restaurant clusters) generate externalities with other firms (i.e., hotels). These industries were chosen due to strong evidence from the hospitality literature indicating that restaurants and hotels affect each other. The restaurant industry is different but related to the hotel industry as restaurant availability is one of the most important considerations when tourists are booking hotel rooms. A link exists between the restaurant and hotel industries that has many similarities with operations and management (e.g., importance of locations, high reliance on labor, service-based businesses), and it provides legitimacy for discussing diverse agglomeration effects in different industries.

To enhance the coherence and rigor of the theoretical foundation, Table 1 provides a structured summary of key studies, categorized by agglomeration types, associated benefits, main findings, and research gaps.

2.2. *Diverse agglomeration effects in the hospitality industry*

Those who have conducted studies on hotel agglomerations, such as Canina et al. (2005), Chung and Kalnins (2001), and Kalnins and Chung (2004), have claimed that hotels are based on demand-side agglomerations. Demand-side agglomeration economies are generated from clustered firms that produce or sell similar products and services. By visiting locations of clustered firms, consumers can compare heterogeneous products and services easily, effectively saving time and reducing costs. According to Canina et al. (2005, p. 567), demand-side benefits stem from "firm heterogeneity that attracts more consumers to an area simply because the area has a wider variety of firms from which to choose." However, studies fail to explain the demand-side benefits of hotels because hotel customers do not visit hotel clusters to compare hotels' different services. Moreover, the theoretical explanations of previous studies align more closely with supply-side agglomeration effects (McCann and Folta, 2009; Woo et al., 2019). For hotels, proximity to locational resources that are not available elsewhere (e.g., the beach, snow, opera, restaurants, museums, or stadiums) is important, as such locations are significant factors that customers consider when choosing hotels (Luo and Yang, 2016). This indicates that the agglomeration of hotels is driven first by the *supply* of important locational resources.

The availability of diverse locational resources is closely related to Jacobs' agglomeration externalities. Because firms gain positive spillover effects due to the diversity of restaurants in cities, hotels will benefit because of diverse consumption options in destinations (Woo et al., 2019). For example, gastronomy tourists or tourists who value special meals during their travel will likely choose hotels that are co-located around good-quality restaurants. Some fine dining

restaurants, such as Michelin-star restaurants, can provide benefits to local hotels: customers can use these restaurants, especially when their hotels don't have high-end F&B services (Leonardi and Moretti, 2023; McCann and Vroom, 2010). Various consumption options from good-quality restaurants increase the awareness and attractiveness of destination areas, which eventually provide positive externalities for hotels (Canina et al., 2005). Given the current trend of hotel firms' strategically focusing on F&B services, hotels are increasingly influenced by the quality of nearby restaurants (Chen and Chang, 2012; Mun et al., 2021; Yeh et al., 2012).

According to previous studies on Jacobs' agglomerations, industries that have relationships with others based on similar production and supply affect each other; furthermore, agglomeration effects become greater if the industrial distance is closer to each other (Duranton and Puga, 2005; Ellison and Glaeser, 1999; Rosenthal and Strange, 2004). This strongly indicates the influence of the positive externalities of restaurant agglomeration on hotels, as both industries have labor-intensive natures that greatly rely on human resources, and they are typically grouped together in the hospitality industry due to similarities in service operations and management. In addition, greater access to skilled workers is crucial for both restaurants and hotels because good-quality human resources, especially service-oriented workers, increase productivity and performance (Lai and Hitchcock, 2017; Mun et al., 2022). The co-location of good-quality restaurants is more likely to provide positive benefits to hotels, as their hospitality-focused workers understand how to provide excellent service (Kadiyali and Kosová, 2013; Kim et al., 2021; Woo and Mun, 2020). This argument aligns with Marshall's supply-side agglomeration perspective, which states that firm agglomeration generates positive externalities due to greater access to specialized labor. From Marshall's and Jacobs' perspectives, this study identifies the positive agglomeration effects of restaurants on hotel performance.

Previous hotel studies have demonstrated positive relationships between agglomeration effects and hotel performance by focusing on demand-side externalities (Canina et al., 2005; Chung and Kalnins, 2001; Kalnins, 2016; Kalnins and Chung, 2004). However, they misused the theoretical mechanism of demand-side agglomerations because they didn't consider the unique characteristics of hotel consumers who are tourists. Tourists may wish to inspect hotel rooms *before* purchasing, but they will visit hotel clusters (i.e., tourist destinations) *after* purchasing hotel rooms. In other words, a reduction in search time and costs would not occur due to the agglomerations of hotels. Even though demand-side externalities may be difficult to generate between hotels, they may occur in industries related to hotels. Tourists purchase multiple products and services while traveling; they purchase not only hotel rooms but also other services, such as flight passes, meals, and tickets to local activities. Unlike other types of products or services, travel requires customers to spend extensive time researching hotels and destinations. From this tourist perspective, hotel consumers will likely choose hotels that are located around a higher concentration of restaurants, as this will help them to reduce the searching time for traveling. Clusters of restaurants are not necessary for hotels but generate demand-side benefits when they are located near hotels.

Hypothesis 1. Co-located restaurants' quality will have positive agglomeration effects on hotel performance, such that hotel performance increases with increases in incumbent restaurants' quality.

2.3. Different beneficiaries' full- vs. limited-service hotels

Previous agglomeration studies have demonstrated positive relationships between agglomeration effects and hotel performance. They also discovered that positive agglomeration effects on hotel performance differ depending on the characteristics of the hotels. For example, hotels receive more benefits if co-located hotels are larger or provide higher-quality services (Canina et al., 2005; Chung and Kalnins, 2001; Jung and Jang, 2019; Kalnins and Chung, 2004). Following previous research, this study assumes that the agglomeration effects of restaurants may influence hotel performance differently based on hotel type.

Table 2
Descriptive statistics.

	Overall	Full-service hotel	Limited-service hotel
ADR	171.474	220.397	146.696
Occupancy rate	76.933	75.557	78.044
RevPAR	133.653	164.854	110.881
Restaurant cluster quality	3.843	3.852	3.834
Full-service	0.508	-	-
Operation	0.396	0.417	0.373
Chain			
Franchised	0.424	0.399	0.449
Independent	0.181	0.184	0.177
Hotel size			
less than 75 rooms	0.065	0.024	0.108
75 – 149 rooms	0.414	0.191	0.648
150 – 299 rooms	0.308	0.399	0.213
300 – 500 rooms	0.130	0.233	0.023
Over 500 rooms	0.082	0.153	0.008
Hotel class			
Economy	0.108	0.002	0.218
Midscale	0.084	0.024	0.146
Upper midscale	0.108	0.034	0.185
Upscale	0.309	0.290	0.329
Upper upscale	0.267	0.437	0.089
Luxury	0.125	0.212	0.033
Hotel age	26.692	31.756	21.390
Observation	10,260	5248	5012

One of the broadest and most prevalent hotel categorizations is the presence of qualified F&B services (Mun et al., 2019; Promsivapallop et al., 2015). When hotels provide a variety of options in terms of F&B services, including restaurants, they are considered full-service hotels. Hotels that provide minimum F&B services are considered limited-service hotels. Restaurant agglomerations may have moderating effects on hotel performance based on the presence of F&B services in hotels. The restaurants in full-service hotels are complementary services of local restaurants, so full-service hotels are partially related to the restaurant industry. Because the restaurant industry is very competitive, full-service hotels must provide similar or even better F&B services to survive in the local market (Mun et al., 2022). Thus, full-service hotels should rely more on restaurants in the area than limited-service hotels.

In other words, full-service hotels would gain Marshall's and Jacobs' agglomerations externalities from clusters of good-quality restaurants as other restaurants. If a local restaurant market has been developed to provide high-quality foods and diverse cuisine, restaurants will likely encounter more foodies and customers with higher standards. To serve and satisfy these customers, businesses look for service-oriented employees and hire people of diverse backgrounds, such as immigrants (Kadiyali and Kosová, 2013). This increases the diversity of employment in an area, and employees can be hired by full-service hotels (Jacobs' agglomeration effects). Thanks to increased diversity in terms of food options and employment in an area, full-service hotels can share inputs used by local restaurants (e.g., seasonal food ingredients and food suppliers) and specialized labor (e.g., star chefs and sommeliers) (Marshall's supply-side agglomeration effects). More importantly, customers who value food, including residents and tourists, are more willing to visit such areas, so full-service hotels have more opportunities to attract customers (Marshall's demand-side agglomeration effects).

On the other hand, since the major focus of limited-service hotels is on rooms rather than F&B, they are less likely to receive agglomeration benefits from co-located restaurants. The most highlighted supply-side benefit in agglomeration studies is knowledge spillover between firms (Woo and Mun, 2020). However, limited-service hotels are less likely to speak to and share information with restaurants. Limited-service hotels may enjoy Jacobs' agglomeration benefits through increased

Table 3
Impact of restaurant cluster quality on nearby hotels' occupancy rates.

	Dependent variable: RevPAR						
	GEE				2SLS		
	Overall		Full-service hotel	Limited-service hotel	Overall	Full-service hotel	Limited-service hotel
Restaurant cluster quality	34.488***	25.041***	41.180***	27.030***	93.920***	113.797***	68.376***
ADR	0.795***	0.793***	0.798***	0.790***	0.777***	0.776***	0.769***
Full-service hotel	−3.913***	−76.340***	-	-	−1.751*	-	-
Full*Restaurant cluster quality	-	18.881***	-	-	-	-	-
Hotel size							
75 – 149 rooms	4.394**	4.369**	17.546***	0.558	2.435	15.594***	−0.563
150 – 299 rooms	2.980	2.976	14.789**	1.543	1.394	13.463**	0.903
300 – 500 rooms	5.333**	5.097**	17.515***	3.223	3.367	15.491***	3.179
Over 500 rooms	7.122**	6.920**	17.931***	6.823*	5.029*	15.327***	5.523
Hotel class							
Midscale	−7.004***	−7.217***	−1.043	−5.820***	−3.824***	−3.033	−2.931**
Upper midscale	−1.999	−1.947	3.928	−0.442	−0.442	2.132	0.962
Upscale	−0.505	−0.418	6.661	1.438	0.216	1.146	2.018
Upper upscale	−3.783**	−3.707**	3.554	−0.455	−2.732	−1.364	0.870
Luxury	−7.510***	−7.280***	0.599	−7.111*	−4.275	−1.885	−2.492
Operating type							
Franchised	3.450***	3.300***	7.357***	−0.750	3.183***	6.666***	−0.451
Independent	−4.817***	−4.709***	−3.325	−4.663**	−2.423	−1.689	−2.195
Hotel age	−0.001	−0.001	0.052	−0.067*	−0.091***	−0.039	−0.141***
Hotel location	Included	Included	Included	Included	Included	Included	Included
County	Included	Included	Included	Included	Included	Included	Included
Constant	−138.030***	−101.919***	−195.156***	−101.087***	−352.275***	−449.313***	−101.087***
R^2	-	-	-	-	0.960	0.966	0.949
Observation	10,260	10,260	5248	5012	8795	4543	4252

Note: the base of operating type was a chain hotel; the base of hotel size was a hotel with less than 75 rooms; the base of hotel class was a economy hotel; *significant at 10 %; **significant at 5 %; ***significant at 1 %; the models used robust standard errors.

employment, but they may not fully use specialized inputs or labor as full-service hotels do. On the other hand, they still receive demand-side agglomeration effects through increased attractiveness and awareness of the area. A cluster of good-quality restaurants attracts more gastronomy tourists, and these customers are likelier to book limited-service hotels if they wish to save money for nicer meals.

Hypothesis 2. The presence of restaurants in hotels positively moderates the link between the quality of co-located restaurants and hotel performance. Full-service hotels achieve better performance than limited-service hotels if they are co-located with better-quality restaurants.

3. Methodology

To investigate the impact of restaurant cluster quality on hotel performance, we measured the scope of agglomeration economies at the county or city level, both of which have been widely used in previous studies (e.g., Canina et al., 2005) and approximate the geographical scope of consumer choice when selecting a restaurant or hotel (Glaeser et al., 2017).

We chose to target California and New York, two popular states for tourists in the United States (US). First, we listed every county or city in both states. Next, we chose 32 counties or cities in which restaurants and hotels are co-located: Albany (e.g., Fig. 1), Anaheim, Bakersfield, Bronx, Lancaster, Long Beach, Long Island, Los Angeles, Monterey Bay, Nevada

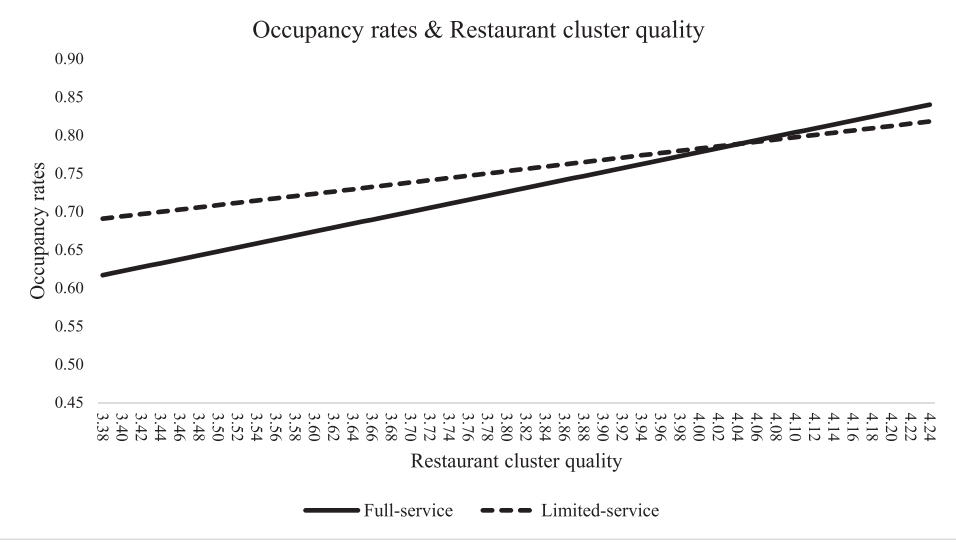


Fig. 2. Full-service hotels' moderating effect on the relationship between restaurant cluster quality and the occupancy rate of nearby hotels (GEE model).

Table 4
Impact of restaurant cluster quality on nearby hotels' ADR.

	Dependent variable: RevPAR						
	GEE				2SLS		
	Overall		Full-service hotel	Limited-service hotel	Overall	Full-service hotel	Limited-service hotel
Restaurant cluster quality	74.431***	44.393***	93.879***	60.968***	315.981***	337.787***	217.564***
Occupancy	2.280***	2.260***	2.741***	1.719***	1.754***	2.206***	1.288***
Full-service hotel	5.715***	−242.081***	-	-	9.185***	-	-
Full*Restaurant cluster quality	-	64.588***	-	-	-	-	-
Hotel size							
75 – 149 rooms	−7.457	−7.362	−14.157	−6.140**	−14.059**	−40.727**	−9.013***
150 – 299 rooms	1.592	1.685	−12.724	3.203	−3.955	−34.164*	0.421
300 – 500 rooms	−9.633	−10.285	−27.478	−4.054	−11.778	−43.011**	−5.738
Over 500 rooms	−21.174***	−21.642***	−46.594***	−27.434***	−15.501**	−52.237***	−23.521***
Hotel class							
Midscale	24.938***	24.056***	22.496**	20.380***	29.938***	36.820*	20.602***
Upper midscale	29.040***	29.111***	21.963*	31.694***	34.251***	36.126*	30.119***
Upscale	43.268***	43.057***	27.024**	48.434***	45.725***	33.881*	46.757***
Upper upscale	60.185***	60.069***	40.341***	70.225***	63.665***	49.215**	67.898***
Luxury	97.179***	97.268***	70.864***	111.143***	113.267***	93.977***	105.923***
Operating type							
Franchised	−21.431***	−21.702***	−46.135***	−0.243	−19.634***	−38.747***	−0.711
Independent	−16.227***	−15.725***	−21.448**	−3.960	−5.066	−9.214	1.891
Hotel age	0.741***	0.732***	0.981***	0.219***	−0.054	−0.022	−0.141**
Hotel location	Included	Included	Included	Included	Included	Included	Included
County	Included	Included	Included	Included	Included	Included	Included
Constant	−388.508***	−261.675***	−438.914***	−295.317***	−1223.034***	−1283.560***	−1030.898***
R^2	-	-	-	-	0.693	0.646	0.776
Observation	10,260	10,260	5248	5012	8759	4543	4252

Note: the base of operating type was a chain hotel; the base of hotel size was a hotel with less than 75 rooms; the base of hotel class was a economy hotel; *significant at 10 %; **significant at 5 %; ***significant at 1 %; the models used robust standard errors.

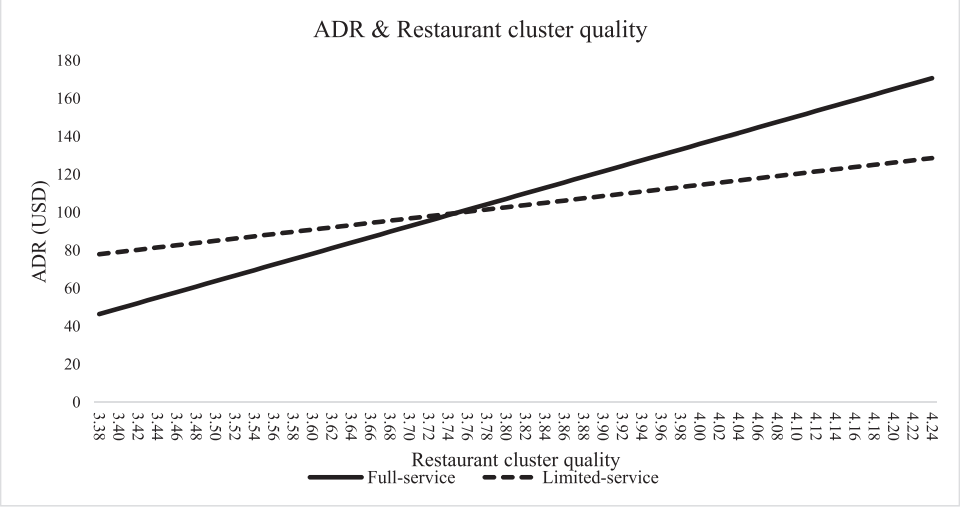


Fig. 3. The moderating effect of full-service hotels on the relationship between restaurant cluster quality and nearby hotels' ADR (GEE model).

City, Newburgh, New Paltz, New York, Oakland, Riverside, Rochester, Sacramento, Schenectady, San Bernardino, San Francisco, Santa Clara, San Louis Obispo, San Diego, San Mateo, Santa Barbara, Santa Cruz, Sutter, Syracuse, Ventura, Yonkers, and Yuba City. Then, we checked whether necessary data were available in each county by matching restaurant data from Yelp and hotel data from STR (formerly Smith Travel Research). Finally, the sample included 21 counties or cities in California and 11 counties or cities in New York, with 10,260 hotel observations and 2923,870 restaurant review ratings.

3.1. Data collection and measurements

To measure the quality of restaurant clusters, we collected data from Yelp, the most popular restaurant review platform. Most restaurants in the US register their businesses on the platform; thus, Yelp effectively represents the business market (Gutt et al., 2019). Yelp's detailed reviews and ratings reflect the extent of service quality and the development of food services in certain areas (Kim et al., 2024). During data collection in April 2023, we first identified every restaurant registered on Yelp in the selected counties or cities. Next, we developed a web crawler to collect all the reviews. From the collected data, we extracted each restaurant's overall rating (1–5 stars), determined monthly ratings based on the reviews' posting dates, and calculated the monthly average restaurant ratings for every selected county or city. Referring to previous agglomeration studies that used Yelp data (e.g., Kim et al., 2024; Liu et al., 2019), we used the monthly average of the overall rating of all

Table 5
Impact of restaurant cluster quality on nearby hotels' RevPAR.

	Dependent variable: RevPAR						
	GEE				2SLS		
	Overall		Full-service hotel	Limited-service hotel	Overall	Full-service hotel	Limited-service hotel
Restaurant cluster quality	133.736***	87.711***	171.816***	96.787***	418.142***	484.170***	33.267***
Full-service hotel	4.511*	−302.532***	-	-	8.345***	-	-
Full*Restaurant cluster quality		80.043***	-	-		-	-
Hotel size							
75 – 149 rooms	−3.807	−3.838	−10.030	−4.944	−13.121**	−33.912*	−10.080**
150 – 299 rooms	0.017	0.021	−11.343	3.938	−6.658	−29.359	−0.644
300 – 500 rooms	−10.974	−11.854*	−22.842	−4.615	−14.107*	−35.911*	−6.175
Over 500 rooms	−22.441***	−22.971***	−41.097**	−20.337***	−18.032**	−46.031**	−19.275**
Hotel class							
Midscale	18.307***	17.332***	35.491**	11.837***	25.511***	45.377*	15.760***
Upper midscale	36.599***	36.391***	53.383***	33.329***	38.065***	57.762**	30.950***
Upscale	51.654***	51.266***	64.520***	51.411***	49.987***	56.021**	48.293***
Upper upscale	67.600***	67.272***	77.602***	71.770***	67.341***	69.496***	68.404***
Luxury	113.695***	113.658***	123.900***	110.734***	124.421***	124.409***	105.536***
Operating type							
Franchised	−21.696***	−22.119***	−35.765***	−4.390	−18.340***	−30.409***	−3.315
Independent	−27.062***	−26.417***	−28.934***	−12.244***	−11.356**	−14.269*	−2.027
Hotel age	0.659***	0.645***	0.854***	0.186***	−0.149**	−0.075	−0.232***
Hotel location	Included	Included	Included	Included	Included	Included	Included
County	Included	Included	Included	Included	Included	Included	Included
Constant	−441.310***	−268.775***	−586.693***	−299.689***	−1475.787***	−1705.720***	−1159.274***
R^2	-	-	-	-	0.673	0.625	0.732
Observation	10,260	10,260	5248	5012	8795	4543	4252

Note: the base of operating type was a chain hotel; the base of hotel size was a hotel with less than 75 rooms; the base of hotel class was a economy hotel; *significant at 10 %; **significant at 5 %; ***significant at 1 %; the models used robust standard errors.

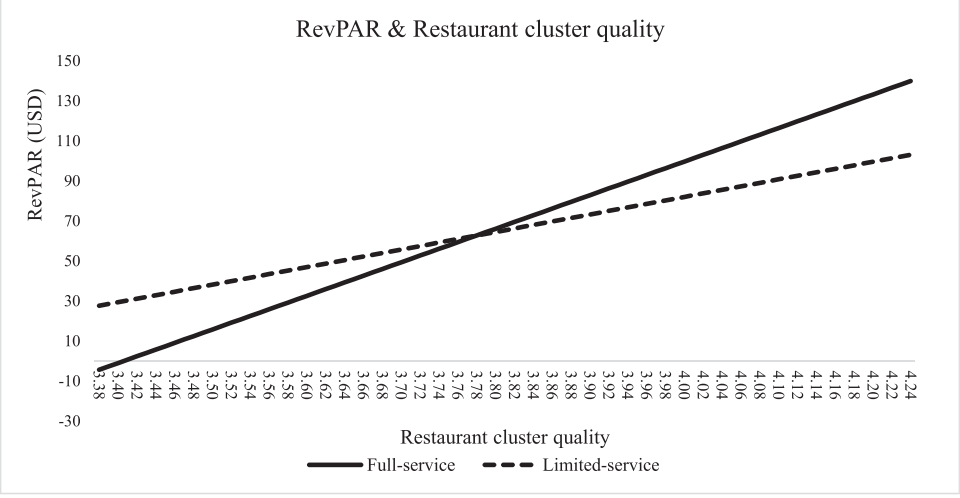


Fig. 4. The moderating effect of full-service hotels on the relationship between restaurant cluster quality and nearby hotels' RevPAR (GEE model).

restaurants in a county or city as a measurement of the quality of a restaurant cluster.

To measure the performance of hotels, we partnered with STR, a well-known company that specializes in managing hotel market data. From the STR reports, we used property-level income statements and connected them with restaurant review rates in their respective counties or cities and years. We included hotel performance data from 2009 to 2019; this specific range was chosen to exclude the extraordinary business circumstances observed during the economic depression in 2008 and the start of the COVID-19 pandemic in 2020. The hotel performance data included information on the occupancy rate, the average daily rate (ADR), and revenue per available room (RevPAR). Based on the information recorded in the STR dataset, we classified the hotels included in this study as either full- or limited-service.

3.2. Statistical models

For statistical analysis, we used county or city controlled generalized estimating equation (GEE) and two-stage least squares (2SLS) panel regression models, both of which use a robust standard error to control the potential heteroskedasticity of the residuals from the fitted model (White, 1980). The GEE models are suitable for longitudinal panel data and reflect correlations between firms (Gardiner et al., 2009; Mun and Woo, 2021). For the 2SLS panel regression models, the length of review and one-year lagged length of review were used as instrument variables to control the endogeneity of restaurant cluster quality. We tested the endogeneity of restaurant cluster quality (e.g., Durbin and Wu-Hauseman), the strength of instrument variables, and over-identifying (e.g., Sargan and Basmann) tests. The test results indicated that the length of review and one-year lagged length of review were



Fig. 5. Summary of findings.

excellent instrument variables because they were significantly related to the restaurant review scores but not significantly related to hotel performance. We compared the statistical results of the two models to ensure the robustness of the findings and the results were consistent.

The restaurant cluster quality (aggregated restaurant review rates) was used as an independent variable, and a hotel's operating performance (occupancy rate, ADR, and RevPAR) was used as a dependent variable. Additionally, the following control variables were used: hotel type (1 for full-service and 0 for limited-service), hotel size (1 for hotels with a small number of rooms and 5 for hotels with a large number of rooms), hotel class (1 for economy and 6 for luxury), hotel operation type (1 for chain, 2 for franchised, and 3 for independent), length of operation (hotel age), hotel location (1 for urban, 2 for suburban, 3 for airport, 4 for interstate/motorway, 5 for resort, and 6 for small metro/town), and hotel county (county or city where a hotel is located).

3.2.1. Model 1

Occupancy rate. The dependent variable (Y_{it}) is RevPAR after controlling for the ADR constant (e.g., since the model included ADR as a control variable, the model represents the impact of Restaurant cluster quality on hotel occupancy instead of RevPAR) such that

$$Y_{it} = \beta_0 + \beta_1 * \text{Restaurant cluster quality}_{it} + \beta_2 * ADR_{it} + \beta_3 * \text{Full-service}_i + \beta_4 * \text{Hotel size}_i + \beta_5 * \text{Hotel class}_i + \beta_6 * \text{Operation}_{it} + \beta_7 * \text{Hotel age}_{it} + \beta_8 * \text{Location}_i + \beta_9 * \text{County}_i + \gamma_{it} + \varepsilon_{it}.$$

Furthermore, the moderating effect of full-service hotels is calculated as follows:

$$Y_{it} = \beta_0 + \beta_1 * \text{Restaurant cluster quality}_{it} + \beta_2 * ADR_{it} + \beta_3 * \text{Full-service}_i + \beta_4 * \text{Quality of restaurant cluster}_{it} * \text{Full-service}_i + \beta_5 * \text{Hotel size}_i + \beta_6 * \text{Hotel class}_i + \beta_7 * \text{Operation}_{it} + \beta_8 * \text{Hotel age}_{it} + \beta_9 * \text{Location}_i + \beta_{10} * \text{County}_i + \gamma_{it} + \varepsilon_{it}.$$

3.2.2. Model 2

ADR. The dependent variable (Y_{it}) is RevPAR after controlling for the occupancy rate constant (e.g., since the model included OCC as a control variable, the model represents the impact of Restaurant cluster quality on hotel ADR instead of RevPAR) such that

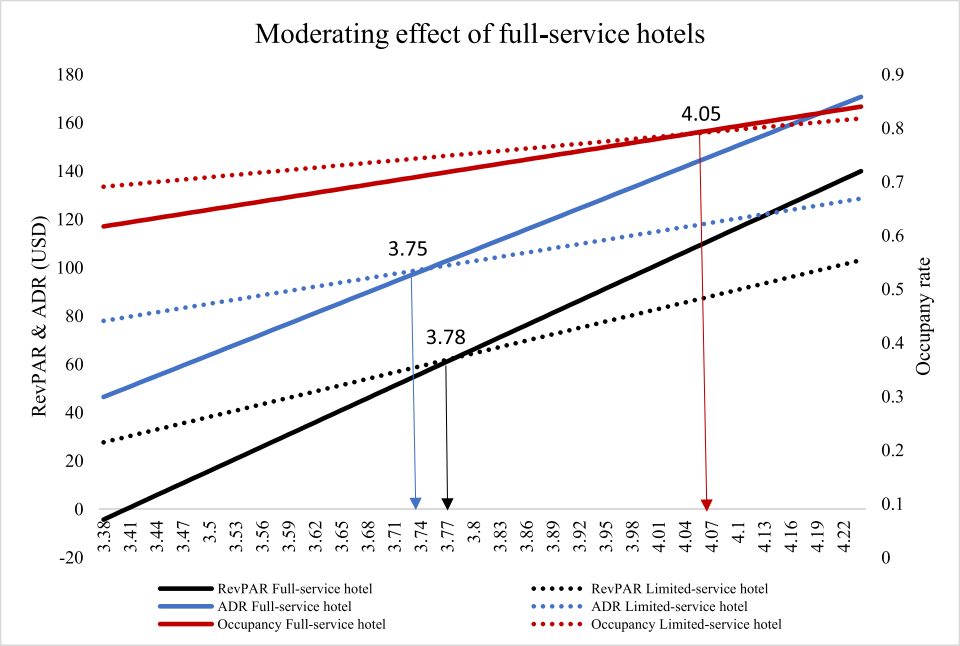


Fig. 6. The moderating effect of full-service hotels on the relationship between restaurant cluster quality and nearby hotels' Occupancy, ADR and RevPAR, ADR (GEE model).

$Y_{it} = \beta_0 + \beta_1 * \text{Restaurant cluster quality}_{it} + \beta_2 * \text{Occupancy rate}_{it} + \beta_3 * \text{Full-service}_i + \beta_4 * \text{Hotel size}_i + \beta_5 * \text{Hotel class}_i + \beta_6 * \text{Operation}_{it} + \beta_7 * \text{Hotel age}_{it} + \beta_8 * \text{Location}_i + \beta_9 * \text{County}_i + \gamma_{it} + \varepsilon_{it}$.

The moderating effect of full-service hotels is calculated as follows:

$Y_{it} = \beta_0 + \beta_1 * \text{Restaurant cluster quality}_{it} + \beta_2 * \text{Occupancy rate}_{it} + \beta_3 * \text{Full-service}_i + \beta_4 * \text{Quality of restaurant cluster}_{it} * \text{Full-service}_i + \beta_5 * \text{Hotel size}_i + \beta_6 * \text{Hotel class}_i + \beta_7 * \text{Operation}_{it} + \beta_8 * \text{Hotel age}_{it} + \beta_9 * \text{Location}_i + \beta_{10} * \text{County}_i + \gamma_{it} + \varepsilon_{it}$.

3.2.3. *Model 3*

RevPAR. The dependent variable (Y_{it}) is RevPAR without controlling for the daily room rate or occupancy rate, such that

$Y_{it} = \beta_0 + \beta_1 * \text{Restaurant cluster quality}_{it} + \beta_2 * \text{Full-service}_{it} + \beta_3 * \text{Hotel size}_i + \beta_4 * \text{Hotel class}_i + \beta_5 * \text{Operation}_{it} + \beta_6 * \text{Hotel age}_{it} + \beta_7 * \text{Location}_i + \beta_8 * \text{County}_i + \gamma_{it} + \varepsilon_{it}$.

The moderating effect of full-service hotels is calculated as follows:

$Y_{it} = \beta_0 + \beta_1 * \text{Restaurant cluster quality}_{it} + \beta_2 * \text{Full-service}_i + \beta_3 * \text{Quality of restaurant cluster}_{it} * \text{Full-service}_i + \beta_4 * \text{Hotel size}_i + \beta_5 * \text{Hotel class}_i + \beta_6 * \text{Operation}_{it} + \beta_7 * \text{Hotel age}_{it} + \beta_8 * \text{Location}_i + \beta_9 * \text{County}_i + \gamma_{it} + \varepsilon_{it}$.

4. Statistical results

As shown in Table 2, ADR and RevPAR values were higher in full-service hotels than in limited-service hotels (ADR; $220 (full-service) vs. $147 (limited-service) and RevPAR; $165 (full-service) vs. $111 (limited-service)). However, the occupancy rate of full-service hotels (76 %) was slightly lower than that of limited-service hotels (78 %). We found that the proportions of full-service hotels and limited-service hotels were similar (51 % and 49 %, respectively). The data also revealed similar proportions of chain and franchised hotels (40 % and 42 %, respectively); independent hotels made up 18 %. Regarding hotel size, 72 % of the hotels included in this study offered 75–299 rooms, and full-service hotels generally comprised more rooms than limited-service hotels. Fifty-seven percent of all hotels were considered upper upscale or upscale, and full-service hotels tended to be classier than limited-service hotels.

Table 3 presents the statistical findings of the GEE and 2SLS models. Overall, the quality of a restaurant cluster had a statistically significant positive impact on nearby hotels' occupancy rates (GEE: $\beta = 34.488$, $p < 0.01$; 2SLS: $\beta = 93.920$, $p < 0.01$). In both models, RevPAR represents the occupancy rates of nearby hotels after controlling for the daily room rate constant. In addition, full-service hotels tended to have lower occupancy rates than limited-service hotels (GEE: $\beta = -3.913$, $p < 0.01$; 2SLS: $\beta = -1.751$, $p < 0.10$), a finding consistent with the descriptive statistics.

The GEE model indicated that full-service hotels have a significantly positive moderating effect on the relationship between a restaurant cluster's quality and the occupancy rates of nearby hotels (GEE: $\beta = 18.881$, $p < 0.01$); the 2SLS models could not include the moderating variable because the models had two instrument variables. Interestingly, hotel size seemed to have a significant positive impact only on full-service hotels' occupancy rates. Hotel class did not consistently impact either full- or limited-service hotels. Fig. 2 shows the positive moderating effect of full-service hotels on the relationship between restaurant cluster quality and hotel occupancy rates within a county or city. That is, as the quality of a restaurant cluster increases, full-service hotels' occupancy rates improve faster than those of limited-service hotels. These results indicate that the quality of the restaurants in a vicinity can motivate tourist demands.

The strength of tourist demands can be indirectly measured by room prices. For example, a hotel may increase room prices when tourist demands are stronger in a destination. As shown in Table 4, in both models, the quality of a restaurant cluster significantly and positively impacts the ADR of nearby hotels (GEE: $\beta = 74.431$, $p < 0.01$; 2SLS: $\beta = 315.981$, $p < 0.01$). Here, RevPAR represents the ADR of nearby hotels after controlling for the occupancy rate constant. Importantly, full-service hotels exhibit a significantly positive moderating effect on the relationship between restaurant cluster quality and nearby hotels' ADR (GEE: $\beta = 64.588$, $p < 0.01$).

Additionally, the models consistently indicated that higher-class hotels charged a higher ADR regardless of hotel service type. In contrast, a hotel's size did not consistently affect its ADR. Fig. 3 clearly shows the positive moderating effect of full-service hotels on the relationship between restaurant cluster quality and nearby hotels' ADR. As seen, full-service hotels can charge more per room than limited-service

hotels because high-quality restaurants in the vicinity can attract more tourists.

The data in Table 5 confirm the impact of restaurant cluster quality on overall hotel performance. If the quality of restaurants in vicinity can induce tourist demand, then the operational performance of nearby hotels will improve substantially due to increased occupancy rates and higher room prices. As expected, restaurant cluster quality had a significant positive impact on the RevPAR of nearby hotels in both models (GEE: $\beta = 133.736$, $p < 0.01$; 2SLS: $\beta = 418.142$, $p < 0.01$). Furthermore, in all models, the RevPAR of nearby hotels was higher among higher-class hotels due to increased room rates.

Consistent with previous findings, we observed that full-service hotels had a significant positive moderating effect on the relationship between restaurant cluster quality and nearby hotels' RevPAR (GEE: $\beta = 80.043$, $p < 0.01$) (Fig. 4). Thus, as review rates improve, the quality of restaurants in a vicinity has a stronger positive impact on full-service hotels' operational performance than on limited-service hotels' performance.

5. Discussion and conclusion

5.1. Summary of findings

Overall, restaurant cluster quality significantly and positively influenced hotel performance in all three models (occupancy, ADR, and RevPAR) even after controlling the endogeneity issues, which strongly supports Hypothesis 1. More specifically, the results of Model 1 prove that restaurant cluster quality attracts tourists to nearby hotels: higher restaurant review rates significantly increase the average occupancy rates of neighboring (within the same county or city) full- and limited-service hotels when other conditions (e.g., room price, hotel class, location, etc.) remain the same. Furthermore, the results of Model 2 confirm that both full- and limited-service hotels within the same county or city can charge tourists higher room prices due to increased customer demands if other conditions (e.g., room occupancy, hotel class, location, etc.) remain constant. These findings support previous research highlighting the role of gourmet restaurants as a pull factor (Daries et al., 2021; De Albuquerque Meneguel et al., 2019) that affects hotel performance positively (e.g., occupancy and ADR). As a result (i.e., due to higher occupancy rates and room prices), both full- and limited-service hotels can enjoy substantial revenue increases (e.g., higher RevPAR) when they are co-located with high-quality restaurants (Model 3).

In addition to the observed positive impacts of high-quality restaurant clusters, we expect to discover further idiosyncratic operational benefits between full- and limited-service hotels. On the one hand, restaurants near full-service hotels can act as business competitors. In this case, full-service hotels compete not only in terms of quality F&B (e.g., taste) and customer service (e.g., employee training) but also in terms of F&B marketing (e.g., pricing) and business strategy (e.g., investments) intended to attract more customers (Mun, Woo, and Paek, 2019). However, such severe competition among full-service hotels and nearby restaurants ultimately improves the quality of F&B offerings in the vicinity as well as full-service hotels' strategic competitiveness compared to limited-service hotels. For example, when NoMad Los Angeles, a small boutique hotel, recently opened its large sitting areas to the public to attract local customers, the locals flocked to and mingled at the hotel bar for after-work drinks, before-dinner aperitifs, and neighborhood nightcaps (Full-Time Travel, 2019). Also, at the NoMad Hotel restaurant in Madison Square Park in New York City, New Yorkers raved about the $78 roasted chicken for two (Polland, 2013).

On the other hand, both full- and limited-service hotels can develop cooperative relationships with nearby restaurants. For example, full-service hotels can take advantage of the same resources as neighboring restaurants, including talented chefs and reliable supply chains for fresh ingredients. For limited-service hotel customers, nearby high-quality restaurants offer dining alternatives with various options. Indeed, Tim Cushman, a James Beard Award-winning celebrity chef and co-founder of the Cushman Concepts restaurant group, developed the F&B program at Park South Hotel in New York City and implemented on-site dining concepts (Demarest, 2019). Similarly, another distinguished Michelin-starred chef, John Fraser, brought an on-site dining experience to guests at LUMA Hotel Times Square in New York City (Demarest, 2019). In practice, few limited-service and full-service hotels have attempted to obtain a synergic operational environment by forming strategic alliances with limited-service hotels and restaurants or by outsourcing hotel F&B operations to third-party F&B companies (Lamminmaki, 2003; Mun, Park, and Woo, 2022). The summary of findings is presented in Fig. 5.

As expected, this study proved that full-service hotels enjoy more operational benefits from high-quality restaurant clusters than limited-service hotels do, which strongly supports Hypothesis 2 (Fig. 6). Our findings also provide evidence of Marshall's and Jacobs' agglomeration externalities in high-quality restaurant clusters. As the quality of restaurants in a vicinity improves (over 4.05), full-service hotels' occupancy rates increase more than limited-service hotels' rates when other conditions (e.g., room price, hotel class, location, etc.) remain constant (Model 1). Model 2's positive moderating effect indicates that full-service hotels charge higher room rates than limited-service hotels as the quality of nearby restaurants improves (over 3.75). Considering the positive relationship between a tourist's income and travel expenditure (Marrocu et al., 2015; Wang et al., 2006), our findings suggest that gourmet tourists may be better able to afford other services at full-service hotels, including pools, spas, gyms, etc., than economical tourists. As a result, full-service hotels experience more agglomeration benefits (RevPAR) from co-located restaurants than limited-service hotels do (Model 3) when the quality of nearby restaurants is higher than 3.78.

5.2. Contributions and implications

One of this study's unique aspects is that we connected the quality of restaurants in a vicinity with tourist destination choices using hotel occupancy rates, room price rates, and revenue as references for tourist demands. Specifically, we tested whether the quality of restaurants near a tourist destination acted as a pull factor by using longitudinal hotel performance data from 2009 to 2019. Previous studies on hotel location selection have highlighted the importance of nearby restaurants, but most of them considered only quantity (i.e., the number of restaurants) or distance (Chou et al., 2008; Yang et al., 2014, 2015). Our research contributes to the literature by suggesting that the quality of nearby restaurants is an important factor influencing hotel performance, including customer demand, room rates, and overall revenue. These findings highlight the importance of incorporating the quality of neighboring restaurants into hotel location selection strategies.

From the viewpoint of agglomeration theory, this study indicates that agglomeration effects can also occur at the inter-industry level. Firms with commonalities and/or complementarities typically agglomerate in certain areas (e.g., Silicon Valley or shopping malls), which affects other firms' locational choices and performance. Existing literature has shown agglomeration effects within industries (i.e., intra-industry spillover) but has yet to focus on agglomeration effects between different industries (Barge-Gil et al., 2020; Kadiyali and Kosová, 2013). Using big data from the restaurant and hotel industries, our findings demonstrate that agglomeration effects can occur between different industries (i.e., inter-industry spillover) without direct interaction or collaboration. This study may serve as an empirical reference for future inter-industry studies on agglomeration effects (Gornig and Schiersch, 2024; Hlee and Lee, 2023).

We tried to reflect the reality and nature of hospitality agglomeration from a tourism perspective and understand various agglomeration effects on hotels using three types of agglomerations: Marshall's supply- and demand-side agglomerations and Jacobs' agglomeration. These

theoretical arguments help confirm the heterogeneous agglomeration effects of the restaurant business on full- and limited-service hotels. Although previous studies on the hospitality and tourism industries have explored agglomeration effects within a specific sector, few have analyzed how these effects vary based on subcategories (e.g., full-service vs limited-service hotels) within the sector (Kim et al., 2021; Li and Liu, 2022; Marco-Lajara et al., 2014). This research contributes to the hospitality and tourism literature on agglomeration effects by proposing an approach that specifies effects at the industry level (Woo and Mun, 2020).

Practically, the findings of this study provide insight into selecting locations for new hotels. First, hotel firms must consider local restaurants' reputations when searching for suitable locations with few hotel-clustered areas. This is because restaurants can significantly influence the attractiveness of a destination as well as compete or cooperate with nearby hotels in food and beverage offerings. Notably, high-quality neighboring restaurants will attract gourmet customers who prefer to stay at full-service hotels rather than at limited-service hotels. Simultaneously, these restaurants compete more directly with full-service hotels in both price and service quality, which also generates stronger positive spillover effects for full-service hotels than for their limited-service counterparts. Second, the positive agglomeration effects of vicinity restaurant quality indicate that hotel managers should proactively consider various collaborating activities with nearby restaurants to improve the quality of their F&B offerings, such as sustainable food sourcing, chef training, and menu development. Thus, not only new but also existing hotels must monitor the quality of the restaurants around them and respond to or reflect changes quickly to sustain business performance. In this way, both hotels and restaurants can enjoy better inter-industry agglomeration effects. Although we acknowledge that other characteristics of restaurants (e.g., cuisine type, menu prices, etc.) can influence hotel performance, we leave these factors for future research, since they were not the focus of this study.

Ethical considerations

There are no human participants in this article and informed consent is not required.

CRediT authorship contribution statement

Seunghun Shin: Writing – review & editing, Methodology, Data curation. **Kim Taekyung:** Writing – review & editing, Resources, Data curation. **Linda Woo:** Writing – review & editing, Validation, Conceptualization. **Mun Sung Gyun:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank STR for providing the data used for the current study.

Statements and declarations

Not applicable.

Data Availability Statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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